

MODULE 1

Claude Foundations & Ecosystem Overview

Complete guide to the Claude model family, product suite, plans, pricing, token economics, safety philosophy, and certification pathway

Claude Certified Architect (CCA-F) | Professional Enterprise Architect Learning Guide | May 2026

What You Will Master in This Module

- Claude model families (Haiku, Sonnet, Opus) — capabilities, benchmarks & selection guide
- Full product suite: Claude.ai, API, Claude Code, Cowork, Chrome/Excel/PowerPoint (beta)
- Plan comparison: Free → Pro → Max → Team → Enterprise — detailed feature matrix
- Current API pricing with prompt caching and batch discount calculations
- Token economics: context windows, input/output ratios, cost modeling
- Safety philosophy: Constitutional AI, Responsible Scaling Policy, ASL levels
- Deployment channels: Direct API, Amazon Bedrock, Google Vertex AI
- CCA-F certification structure, exam domains, and Skilljar learning path

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1.1 What Is Claude & Anthropic's Mission

Claude is Anthropic's family of large language models, built with a foundational emphasis on safety, helpfulness, and honesty. Anthropic is an AI safety company, and that mission shapes every product decision — from training methodology to deployment policy.

For enterprise architects, understanding Anthropic's safety-first stance is not optional: it directly determines what Claude can and cannot do in your application, which behaviors operators can customize, and what compliance assurances you can make to regulators.

The Three Core Properties (Always Tested in Domain 5)

Property	What It Means in Practice
Helpful	Genuinely assists users and operators; follows instructions precisely; produces high-quality outputs without unnecessary
Harmless	Avoids producing content that is dangerous, misleading, illegal, or harmful. Trained via Constitutional AI to have safety l
Honest	Does not deceive users or operators; expresses uncertainty appropriately; resists sycophantic pressure to agree when v

Architect Insight: These three properties create real architectural constraints. 'Helpful' means your system prompt shouldn't over-restrict Claude unnecessarily. 'Harmless' means you cannot use system prompts to bypass safety training. 'Honest' means Claude will not lie to users on your behalf — design your application accordingly.

1.2 Model Families: Haiku, Sonnet, Opus

The Claude model family follows a consistent three-tier structure. Each tier is updated through version generations (4, 4.1, 4.5, 4.6, 4.7, 4.8...). The version number and date suffix in the API model string uniquely identify a model snapshot, ensuring reproducibility in production.

Model	API String	Profile	Best Use Cases	\$/MTok in
Claude Haiku 4.5	claude-haiku-4-5-20251001	Fastest & most cost-efficient	High-volume routing, classification, quick QA, sub-agent worker tasks,	\$1.085
Claude Sonnet 4.6	claude-sonnet-4-6	Best balanced (default)	Production workloads: code generation, document analysis, customer s	\$3.1015
Claude Opus 4.7	claude-opus-4-7	Most capable (flagship)	Complex reasoning, autonomous coding, 128K context window, 20-step agentic tasks, hard	\$5.1025
Claude Opus 4.8	claude-opus-4-8	Latest frontier	Hardest coding/reasoning tasks; new to 150K context window (up to 35% more tokens	\$5.1025 (up to)

Context Window: All current Opus and Sonnet models support a **1 million token** context window at standard pricing — no surcharge. Haiku 4.5 supports 200K tokens. This is approximately 750,000 words or an entire medium-sized codebase.

Model Selection Decision Tree

Requirement	Model Choice
Need	→ Choose

Sub-500ms latency + high volume (100K+ req/day)	→	Haiku 4.5
General production API workload (default choice)	→	Sonnet 4.6
Complex multi-step reasoning, hard coding tasks	→	Opus 4.7 or 4.8
Budget-constrained high-volume analysis	→	Haiku 4.5 + Batch API
Research / evaluation / maximum quality	→	Opus 4.8 + Extended Thinking
Cost-sensitive production with long static prompts	→	Any model + Prompt Caching

Exam Note: The CCA-F exam references **claude-sonnet-4-20250514** in code examples. Real-world model strings use the format: *claude-{tier}-{major}.{minor}-{YYYYMMDD}* or abbreviated forms. Always check the Anthropic models documentation for the latest stable string before deploying.

Important: Opus 4.8 Tokenizer Change

Claude Opus 4.8 ships with a new tokenizer that can generate up to **35% more tokens** for the same input text compared to Opus 4.6. Per-token prices are unchanged, but effective cost per request can increase by up to 35%. Always benchmark your specific workload before migrating from Opus 4.6 to 4.8.

1.3 The Claude Product Suite in Depth

Claude is not a single product. It is an ecosystem of access channels, each optimized for a different user persona and integration pattern. Enterprise architects must understand all channels to design the right deployment architecture.

Product	Access Type	Primary Users	Key Capabilities
Claude.ai	Web / Mobile / Desktop	End users, knowledge workers	Onboard with Projects, file uploads, memory, MCP connectors. Free, Pro, Enterprise
Claude API	REST API + Python/TS SDKs	Developers	Full programmatic access. Messages API, tool use, batch, streaming, e
Claude Code	CLI + Agent SDK	Software engineers	Agentic coding: autonomous file editing, shell commands, MCP tools, h
Claude Cowork	Desktop GUI	Non-technical professional	File management, task automation, workflow orchestration — no code r
Claude in Chrome	Browser extension (beta)	Power users	Browsing agent: reads page content, clicks, fills forms, executes multi-s
Claude in Excel	Excel add-in (beta)	Analysts / Finance	Formula generation, data analysis, chart creation, pivot automation with
Claude in PowerPoint	PPT add-in (beta)	Presenters	Slide creation, design suggestions, content generation inside Microsoft
Amazon Bedrock	AWS-managed API	AWS enterprise teams	Bedrock converse API, IAM auth, VPC integration, private link, data res
Google Vertex AI	GCP-managed API	GCP enterprise teams	Vertex AI Claude endpoint, CMEK, VPC service controls, regional endp
Anthropic Academy	Self-serve LMS	Architects / developers	Official courses: Claude 101, API, Bedrock, MCP, Claude Code, Skills,

1.4 Subscription Plan Comparison

Claude.ai offers consumer and enterprise plans. API access is billed separately per token. Understanding plan differences matters for architects designing who accesses Claude via the UI vs. building custom API-powered applications.

Feature	Free	Pro \$20/mo	Max 5x \$100/mo	Max 20x \$200/mo	Team Std \$25/seat	Enterprise Custom
Message limits	Low daily	~5x Free	~25x Free	~100x Free	Higher	Negotiated
Model access	Sonnet	Opus+Sonnet	Opus+Sonnet	Opus+Sonnet+Priority	Opus+Sonnet	All incl. Opus 4.8
Extended thinking	No	Yes	Yes	Yes	Yes	Yes
Projects	Limited	Unlimited	Unlimited	Unlimited	Unlimited	Unlimited + admin
Memory (auto)	No	Beta	Beta	Beta	Beta	Yes + private data
MCP connectors	No	No	No	No	Yes	Yes + custom MCPs
Claude Code usage	No	No	Max-scale	Max-scale	Team Claude Code	Enterprise Claude Code
SSO / SAML	No	No	No	No	No	Yes
Audit logs	No	No	No	No	Limited	Full + SIEM export
Zero data retention	No	No	No	No	No	Optional
SLA	No	No	No	No	No	Yes
Priority support	No	No	No	No	No	Yes + CSM
Admin console	No	No	No	No	Basic	Full governance

Architect Decision: For enterprise deployments, choose **Enterprise plan for UI users** (SSO, audit logs, admin control) and use the **API directly or via Bedrock/Vertex** for programmatic access. Do not rely on Claude.ai plans for production API workloads — use the API with token-based billing for predictable cost control.

1.5 API Pricing & Cost Modeling

API pricing is per million tokens (MTok), billed separately for input and output tokens. The output cost is consistently 5x the input cost across all current tiers. Two major discount levers — prompt caching and batch processing — can dramatically reduce costs in production pipelines.

Model	Input \$/MTok	Output \$/MTok	Cache Write \$/MTok	Cache Read \$/MTok	Batch Input \$/MTok	Batch Output \$/MTok
Haiku 4.5	\$1.00	\$5.00	\$1.25	\$0.10	\$0.50	\$2.50
Sonnet 4.6	\$3.00	\$15.00	\$3.75	\$0.30	\$1.50	\$7.50
Opus 4.7 / 4.8	\$5.00	\$25.00	\$6.25	\$0.50	\$2.50	\$12.50
Opus 4.6 Fast Mode	\$30.00	\$150.00	N/A	N/A	N/A	N/A

Cost Modeling Examples

Use these formulas to budget your Claude API spend before production launch:

```
# Example 1: Standard Sonnet 4.6 document analysis # Input: 10,000 tokens system+document, Output: 500 tokens summary
cost_per_req = (10_000 * 3.00 / 1_000_000) + (500 * 15.00 / 1_000_000) # = $0.03 + $0.0075 = $0.0375 per request
# Example 2: Same with prompt caching (system prompt cached = 2,000 tokens)
cost_cached = (2_000 * 0.30 / 1_000_000) + # cache read (8_000 * 3.00 / 1_000_000) + # fresh input (500 * 15.00 / 1_000_000)
# output # = $0.0006 + $0.024 + $0.0075 = $0.0321 — 14% savings
# Example 3: Batch API for 10,000 documents nightly
batch_cost = 10_000 * ((10_000 * 1.50 / 1_000_000) + (500 * 7.50 / 1_000_000)) # = $187.50 vs $375 standard — 50% savings = $187.50 saved per run
```

Cost Optimization Priority Order: (1) Use Haiku 4.5 for tasks where quality difference is negligible. (2) Add prompt caching for any static system prompt > 1,024 tokens. (3) Switch non-real-time bulk workloads to Batch API. (4) Route requests by complexity — Haiku for simple, Sonnet for medium, Opus for hard. Combined, these can reduce costs by 70-90%.

1.6 Token Economics & Context Windows

Tokens are the fundamental unit of measurement for all Claude API interactions. A token is roughly 4 characters or 0.75 words in English. Different languages have different token densities — code, JSON, and non-Latin scripts often tokenize differently.

1 token ≈	4 characters or ~0.75 words in English text
1,000 tokens ≈	750 words — about 3 pages of double-spaced text
Max context (Opus/Sonnet 4.6+)	1,000,000 tokens — ~750,000 words, ~2,000 pages
Max context (Haiku 4.5)	200,000 tokens — ~150,000 words, ~500 pages
Max output (Opus 4.6/4.7/4.8)	128,000 tokens — extended with output-300k-2026-03-24 beta to 300K
Max output (Sonnet 4.6)	64,000 tokens; 300K with beta header
Max output (Haiku 4.5)	8,192 tokens standard
Token counting API	Use /v1/messages/count_tokens to pre-calculate before billing
Context auto-compaction	Claude Code compacts context at ~75-92% capacity — summarizes history
Cache minimum	1,024 tokens minimum to be eligible for prompt caching
Thinking tokens	Extended thinking tokens count toward max_tokens but previous thinking blocks are stripped

Context Window Strategy for Architects

Scenario	Architectural Recommendation
Static system prompt	Mark with cache_control: ephemeral — free cache refresh after 5 min, ~90% discount on hits
Large reference document	Chunk it — or cache the document in the context for repeated queries
Long conversation history	Implement server-side summarization at ~70% capacity; never rely on auto-compaction for API apps

RAG retrieved chunks	Inject at the end of the user message — after the static cached content
Multi-modal inputs	Images cost ~1,600 tokens each regardless of image size; PDFs counted per page
Measuring actual usage	Always log response.usage.input_tokens + output_tokens for cost tracking

1.7 Constitutional AI & Safety Philosophy

Constitutional AI (CAI) is Anthropic's training methodology that embeds safety behaviors directly into the model rather than applying them as post-hoc filters. Understanding CAI is essential for Domain 5 of the CCA-F exam and for designing compliant enterprise applications.

How Constitutional AI Works

Phase	Description
1. RLHF (Base)	Standard reinforcement learning from human feedback to create a capable base model
2. Constitution	A set of principles ('the constitution') defining what makes Claude's responses good or bad
3. AI Feedback (RLAIF)	Claude critiques its own outputs against the constitution — no human labeling needed at scale
4. Revision	Claude revises responses based on its own constitutional critique
5. Fine-tuning	The revised outputs train the final model — safety behaviors are intrinsic, not a filter

The Three-Tier Permission System

Claude's behavior is governed by a three-tier system. Each tier can only operate within the bounds set by the tier above it:

Tier	Authority Level	What They Control
TIER 1 Anthropic (Training)	Highest authority	Sets absolute limits via Constitutional AI training. Cannot be overridden by any system prompt, user
TIER 2 Operators (System Prompt)	Middle authority	Can adjust Claude's default behaviors within Anthropic's usage policy. Cannot exceed Tier 1 limits. C
TIER 3 Users (Chat Messages)	Lowest authority	Can adjust behaviors within what operators permit. Cannot exceed Tier 2 limits. Examples: request o

Hard Limits (Cannot Be Overridden at Any Tier)

- Generating sexual content involving minors (CSAM) — absolute, no exceptions
- Providing meaningful technical uplift for creating CBRN (chemical, biological, radiological, nuclear) weapons
- Creating functional cyberweapons or malware designed to cause significant damage

- Denying being an AI when a user sincerely and directly asks
- Taking actions that undermine the ability of humans to oversee and correct AI systems
- Assisting attempts to seize unprecedented societal control

Adjustable Default Behaviors (Operator-Configurable)

Behavior Category	Examples
Default ON — operators CAN turn off	Safe Messaging guidelines for self-harm/suicide; adding safety caveats; balanced perspectives on c
Default OFF — operators CAN turn on	Explicit Adult content (age-appropriate platforms); detailed drug harm reduction without warnings; tak
Default ON — users CAN turn off	Adding disclaimers to persuasive essays; suggesting professional help for personal struggles; break
Default OFF — users CAN turn on	Using crude/profane language; providing extremely blunt feedback without diplomatic softening

1.8 Responsible Scaling Policy (RSP)

The Responsible Scaling Policy establishes the framework by which Anthropic decides when to train and deploy increasingly powerful models. It defines AI Safety Levels (ASL) that trigger specific safety requirements. Enterprise architects should understand this because it explains why certain capabilities are gated or restricted.

Level	Threshold	Implications
ASL-1	Current commercial models	No extraordinary risks identified. Standard deployment permitted.
ASL-2	Models approaching serious uplift for CBRN or cyber capabilities	Enhanced evaluations required. Standard deployment with monitoring.
ASL-3	Models capable of serious CBRN uplift or autonomous operations	Autonomous operations required before deployment. Not yet reached.
ASL-4	Models posing catastrophic risk	Full capability control protocols required. Theoretical at this stage.

Enterprise Implication: The RSP means that Anthropic proactively withholds certain capabilities if safety evaluations are not met — even if those capabilities are technically achievable. For example, Claude Mythos Preview (the most advanced frontier model) is restricted to trusted organizations via Project Glasswing due to cybersecurity concerns.

1.9 Deployment Channels: Direct API, Bedrock, Vertex

Enterprise architects must choose the right deployment channel based on infrastructure, compliance, data residency, and operational requirements. Each channel offers different SLAs, IAM integrations, and data isolation guarantees.

Feature	Direct Anthropic API	Amazon Bedrock	Google Vertex AI
Authentication	ANTHROPIC_API_KEY	AWS IAM roles + Cognito	GCP service accounts + OAuth
Network isolation	Internet (TLS)	VPC endpoint + PrivateLink	VPC Service Controls
Data residency	Anthropic infrastructure	Your AWS account + region	Your GCP project + region

Anthropic access to data	Logging per retention policy	None — your VPC	None — your GCP project
Compliance certs	SOC 2 Type II	HIPAA, SOC 2, FedRAMP	HIPAA, SOC 2, ISO 27001
Cost model	Per-token billing	Per-token + Bedrock inference markup	
Per-token + Vertex markup			
Prompt caching	Full support	Full support	Full support
Batch API	Yes	Yes (Batch Inference)	Yes (Batch predictions)
Extended thinking	Yes	Yes (beta headers)	Yes (beta headers)
Rate limits	Anthropic tiers	Bedrock service quotas	Vertex quotas
Best for	Startups, speed to market	AWS-native enterprise	GCP-native enterprise

Bedrock SDK Pattern — Python:

```
import boto3, json
bedrock = boto3.client('bedrock-runtime', region_name='us-east-1')
response = bedrock.invoke_model(modelId='anthropic.claude-sonnet-4-6-20251101-v1:0', body=json.dumps({'anthropic_version': 'bedrock-2023-05-31', 'max_tokens': 1024, 'messages': [{'role': 'user', 'content': 'Hello from Bedrock'}]}))
result = json.loads(response['body'].read())
print(result['content'][0]['text'])
```

1.10 CCA-F Certification Pathway

The Claude Certified Architect — Foundations (CCA-F) is Anthropic's official certification for practitioners who design and build production Claude applications. It was launched in late 2025 through Anthropic Academy on the Skilljar platform.

Exam Code	CCA-F
Questions	60 multiple-choice
Duration	120 minutes
Passing Score	720 / 1,000 (approximately 72% — weighted by domain)
Proctoring	Online via Skilljar — webcam + screen share required
Retake Policy	14-day waiting period after a failed attempt
Validity	2 years from pass date
Prerequisites (recommended)	3–6 months hands-on Claude development experience
Target Candidate	Solution architect designing/shipping production Claude applications
Platform	anthropic.skilljar.com — Anthropic Academy

Exam Domain Weights — Know These Cold

Domain	Topic	Weight	Study Strategy
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Domain 1	Claude API & SDK Integration	22%	Priority 1 — most questions; master Messages API, streaming, tool use, ca
Domain 2	Prompt Engineering & Optimization	20%	Priority 2 — XML tags, few-shot, chain-of-thought, extended thinking, RAG
Domain 3	Model Context Protocol (MCP)	18%	Priority 3 — hardest for most; tools/resources/prompts primitives, transports
Domain 4	Agent Design & Orchestration	22%	Priority 1 — tied with Domain 1; CLAUDE.md, Skills, Subagents, multi-agen
Domain 5	Safety, Security & Responsible Deployment	18%	Priority 3 — frequently underestimated; hard limits, privacy, hardening, RSP

Recommended Skilljar Course Completion Order

#	Course	Level	Why It Matters
1	Claude 101	Beginner	Start here — core features, projects, file uploads, everyday Claude usage
2	Building with the Claude API	Intermediate	Most important course — covers full API from auth to agents
3	Claude in Amazon Bedrock	Intermediate	AWS-specific deployment, IAM, VPC, RAG on Bedrock
4	Introduction to Model Context Protocol	Intermediate	Three MCP primitives, Python SDK, MCP Inspector
5	MCP: Advanced Topics	Advanced	Sampling, transport layers, roots, production scaling
6	Claude Code in Action	Intermediate	Agentic coding, CLAUDE.md, GitHub workflows, thinking modes
7	Introduction to Agent Skills	Intermediate	SKILL.md frontmatter, description writing, distribution
8	Introduction to Subagents	Intermediate	Isolated contexts, delegation, structured outputs, limits